Topic: The Dirac delta function

Question: Solve the initial value problem and find the general solution, given $y(0) = 0$ and $y'(0) = -2$.

$$y'' + 3y' + 2y = 7\delta(t - 9)$$

Answer choices:

A $y(t) = 7e^{9-t}u(t-9) + 2e^{-3t}$

B $y(t) = 7e^{9-t}u(t-9) - 2e^{-3t}$

C $y(t) = 7e^{9-t}u(t-9) + 7e^{18-2t}u(t-9) + 2e^{-t} - 2e^{-2t}$

D $y(t) = 7e^{9-t}u(t-9) - 7e^{18-2t}u(t-9) - 2e^{-t} + 2e^{-2t}$

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Solution: D

Apply the Laplace transform to both sides of the second order equation.

$$\mathscr{L}(y'') + 3\mathscr{L}(y') + 2\mathscr{L}(y) = 7\mathscr{L}(\delta(t-9))$$

Using formulas from our table of transforms,

$$\mathscr{L}(y''(t)) = s^2Y(s) - sy(0) - y'(0)$$

$$\mathscr{L}(y'(t)) = sY(s) - y(0)$$

$$\mathscr{L}(y(t)) = Y(s)$$

$$\mathscr{L}(\delta(t-c)) = e^{-cs}$$

we get

$$s^2Y(s) - sy(0) - y'(0) + 3(sY(s) - y(0)) + 2Y(s) = 7e^{-9s}$$

Substitute the initial conditions $y(0) = 0$ and $y'(0) = -2$.

$$s^2Y(s) - s(0) - (-2) + 3sY(s) - 3(0) + 2Y(s) = 7e^{-9s}$$

$$s^2Y(s) + 2 + 3sY(s) + 2Y(s) = 7e^{-9s}$$

Solve for $Y(s)$.

$$(s^2 + 3s + 2)Y(s) = 7e^{-9s} - 2$$

$$Y(s) = \frac{7}{s^2 + 3s + 2}e^{-9s} - \frac{2}{s^2 + 3s + 2}$$

$$Y(s) = 7\frac{1}{(s+2)(s+1)}e^{-9s} - 2\frac{1}{(s+2)(s+1)}$$

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Apply a partial fractions decomposition.

$$\frac{1}{(s+2)(s+1)} = \frac{A}{s+2} + \frac{B}{s+1}$$

To solve for A, remove the $s+2$ factor from the denominator of the left side, then substitute $s = -2$ into the left side.

$$\frac{1}{s+1} \to \frac{1}{-2+1} \to \frac{1}{-1} \to -1$$

To solve for B, remove the $s+1$ factor from the denominator of the left side, then substitute $s = -1$ into the left side.

$$\frac{1}{s+2} \to \frac{1}{-1+2} \to \frac{1}{1} \to 1$$

So the decomposition is

$$\frac{1}{(s+2)(s+1)} = \frac{-1}{s+2} + \frac{1}{s+1}$$

$$\frac{1}{(s+2)(s+1)} = \frac{1}{s+1} - \frac{1}{s+2}$$

Then the Laplace transform becomes

$$Y(s) = 7\left(\frac{1}{s+1} - \frac{1}{s+2}\right)e^{-9s} - 2\left(\frac{1}{s+1} - \frac{1}{s+2}\right)$$

$$Y(s) = 7\left(\frac{1}{s+1}\right)e^{-9s} - 7\left(\frac{1}{s+2}\right)e^{-9s} - 2\left(\frac{1}{s+1}\right) + 2\left(\frac{1}{s+2}\right)$$

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Now that we've broken down the transform this way, we can apply the inverse transform to find the general solution to the second order nonhomogeneous equation.

$$y(t) = 7(e^{-(t-9)})u(t-9) - 7(e^{-2(t-9)})u(t-9) - 2e^{-t} + 2e^{-2t}$$

$$y(t) = 7e^{9-t}u(t-9) - 7e^{18-2t}u(t-9) - 2e^{-t} + 2e^{-2t}$$

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Topic: The Dirac delta function

Question: Solve the initial value problem and find the general solution, given $y(0) = 0$ and $y'(0) = 0$.

$$y'' + 5y = 2\delta(t-2) + u(t-2)$$

Answer choices:

A $y(t) = \frac{1}{5}u(t-2)(\sin(\sqrt{5}(t-2)) - \cos(\sqrt{5}(t-2)))$

B $y(t) = \frac{1}{5}u(t-2)(2\sqrt{5}\sin(\sqrt{5}(t-2)) - \cos(\sqrt{5}(t-2)) + 1)$

C $y(t) = \frac{1}{5}u(t-2)(\sin(\sqrt{5}(t-2)) + \cos(\sqrt{5}(t-2)))$

D $y(t) = \frac{1}{5}u(t-2)(2\sqrt{5}\sin(\sqrt{5}(t-2)) + \cos(\sqrt{5}(t-2)) - 1)$

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Solution: B

Apply the Laplace transform to both sides of the second order equation.

$$\mathscr{L}(y'') + 5\mathscr{L}(y) = 2\mathscr{L}(\delta(t-2)) + \mathscr{L}(u(t-2))$$

Using formulas from our table of transforms,

$$\mathscr{L}(y''(t)) = s^2Y(s) - sy(0) - y'(0)$$

$$\mathscr{L}(y(t)) = Y(s)$$

$$\mathscr{L}(\delta(t-c)) = e^{-cs}$$

$$\mathscr{L}(u(t-c)) = \frac{e^{-cs}}{s}$$

we get

$$s^2Y(s) - sy(0) - y'(0) + 5Y(s) = 2e^{-2s} + \frac{e^{-2s}}{s}$$

Substitute the initial conditions $y(0) = 0$ and $y'(0) = 0$.

$$s^2Y(s) - s(0) - 0 + 5Y(s) = 2e^{-2s} + \frac{e^{-2s}}{s}$$

$$s^2Y(s) + 5Y(s) = 2e^{-2s} + \frac{e^{-2s}}{s}$$

Solve for $Y(s)$.

$$(s^2 + 5)Y(s) = 2e^{-2s} + \frac{e^{-2s}}{s}$$

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$$Y(s) = \frac{2e^{-2s}}{s^2+5} + \frac{e^{-2s}}{s(s^2+5)}$$

$$Y(s) = 2\left(\frac{1}{s^2+5}\right)e^{-2s} + \frac{1}{s(s^2+5)}e^{-2s}$$

Apply a partial fractions decomposition.

$$\frac{1}{s(s^2+5)} = \frac{A}{s} + \frac{Bs+C}{s^2+5}$$

Expand the decomposition,

$$1 = A(s^2+5) + (Bs+C)s$$

$$1 = As^2 + 5A + Bs^2 + Cs$$

$$1 = (A+B)s^2 + (C)s + (5A)$$

then equate coefficients.

$$A + B = 0$$

$$C = 0$$

$$5A = 1$$

From these equations, we see $A = 1/5$, and then $B = -1/5$. Therefore, the decomposition is

$$\frac{1}{s(s^2+5)} = \frac{\frac{1}{5}}{s} + \frac{-\frac{1}{5}s + 0}{s^2+5}$$

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$$\frac{1}{s(s^2+5)} = \frac{1}{5}\left(\frac{1}{s}\right) - \frac{1}{5}\left(\frac{s}{s^2+5}\right)$$

Then the Laplace transform can be rewritten as

$$Y(s) = 2\left(\frac{1}{s^2+5}\right)e^{-2s} + \left(\frac{1}{5}\left(\frac{1}{s}\right) - \frac{1}{5}\left(\frac{s}{s^2+5}\right)\right)e^{-2s}$$

$$Y(s) = \frac{2\sqrt{5}}{5}\left(\frac{\sqrt{5}}{s^2+5}\right)e^{-2s} + \frac{1}{5}\left(\frac{1}{s}\right)e^{-2s} - \frac{1}{5}\left(\frac{s}{s^2+5}\right)e^{-2s}$$

Now that we've broken down the transform this way, we can apply the inverse transform to find the general solution to the second order nonhomogeneous equation.

$$y(t) = \frac{2\sqrt{5}}{5}\sin(\sqrt{5}(t-2))u(t-2) + \frac{1}{5}u(t-2) - \frac{1}{5}\cos(\sqrt{5}(t-2))u(t-2)$$

$$y(t) = \frac{1}{5}u(t-2)(2\sqrt{5}\sin(\sqrt{5}(t-2)) - \cos(\sqrt{5}(t-2)) + 1)$$

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Topic: The Dirac delta function

Question: Solve the initial value problem and find the general solution, given $y(0) = 0$ and $y'(0) = 3$.

$$y'' + y' = \delta(t - 3)$$

Answer choices:

A $y(t) = u(t-3) - e^{3-t}u(t-3) - e^{-3t}$

B $y(t) = u(t-3) + e^{-t}u(t-3) + 3e^{-t} + 3$

C $y(t) = u(t-3) - u(t-3)e^{-(t-3)} + 3 - 3e^{-t}$

D $y(t) = u(t-3) - e^{-t}u(t-3) + e^{-3t}$

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Solution: C

Apply the Laplace transform to both sides of the second order equation.

$$\mathscr{L}(y'') + \mathscr{L}(y') = \mathscr{L}(\delta(t-3))$$

Using formulas from our table of transforms,

$$\mathscr{L}(y''(t)) = s^2Y(s) - sy(0) - y'(0)$$

$$\mathscr{L}(y'(t)) = sY(s) - y(0)$$

$$\mathscr{L}(y(t)) = Y(s)$$

$$\mathscr{L}(\delta(t-c)) = e^{-cs}$$

we get

$$s^2Y(s) - sy(0) - y'(0) + (sY(s) - y(0)) = e^{-3s}$$

Substitute the initial conditions $y(0) = 0$ and $y'(0) = 3$.

$$s^2Y(s) - s(0) - 3 + sY(s) - 0 = e^{-3s}$$

$$s^2Y(s) - 3 + sY(s) = e^{-3s}$$

Solve for $Y(s)$.

$$(s^2 + s)Y(s) = e^{-3s} + 3$$

$$Y(s) = e^{-3s}\frac{1}{s^2+s} + \frac{3}{s^2+s}$$

$$Y(s) = e^{-3s}\left(\frac{1}{s(s+1)}\right) + 3\left(\frac{1}{s(s+1)}\right)$$

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Apply a partial fractions decomposition.

$$\frac{1}{s(s+1)} = \frac{A}{s} + \frac{B}{s+1}$$

To solve for A, remove the s factor from the denominator of the left side, then substitute $s = 0$ into the left side.

$$\frac{1}{s+1} \to \frac{1}{0+1} \to \frac{1}{1} \to 1$$

To solve for B, remove the $s+1$ factor from the denominator of the left side, then substitute $s = -1$ into the left side.

$$\frac{1}{s} \to \frac{1}{-1} \to -1$$

So the decomposition is

$$\frac{1}{s(s+1)} = \frac{1}{s} + \frac{-1}{s+1}$$

$$\frac{1}{s(s+1)} = \frac{1}{s} - \frac{1}{s+1}$$

Then the Laplace transform becomes

$$Y(s) = e^{-3s}\left(\frac{1}{s} - \frac{1}{s+1}\right) + 3\left(\frac{1}{s} - \frac{1}{s+1}\right)$$

$$Y(s) = e^{-3s}\left(\frac{1}{s}\right) - e^{-3s}\left(\frac{1}{s+1}\right) + 3\left(\frac{1}{s}\right) - 3\left(\frac{1}{s+1}\right)$$

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Now that we've broken down the transform this way, we can apply the inverse transform to find the general solution to the second order nonhomogeneous equation.

$$y(t) = u(t-3) - u(t-3)e^{-(t-3)} + 3 - 3e^{-t}$$

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